

19. Neural Processing Unit (NPU)

19.1 Overview

The MCU provides the Arm® Ethos™-U55 NPU.

[Table 19.1](#) lists the NPU specifications.

Table 19.1 NPU specifications

Parameter	Specifications
Number of 8x8 MACs	256 units
Network	8-bit and 16bit integer quantized Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN)
Compression	8-bit weights
DMAC	The NPU includes a DMAC. The DMAC transfer the following data. • The command stream • Network weights • Bias information • Scale information • Input Feature Maps (IFMs) • Output Feature Maps (OFMs)
Interrupt	NPU_IRQ
TrustZone filter	Security and Privilege attribution can be changed by the Register RESET in NPU. Initial value is set by OFS2

[Figure 19.1](#) shows the NPU block diagram.

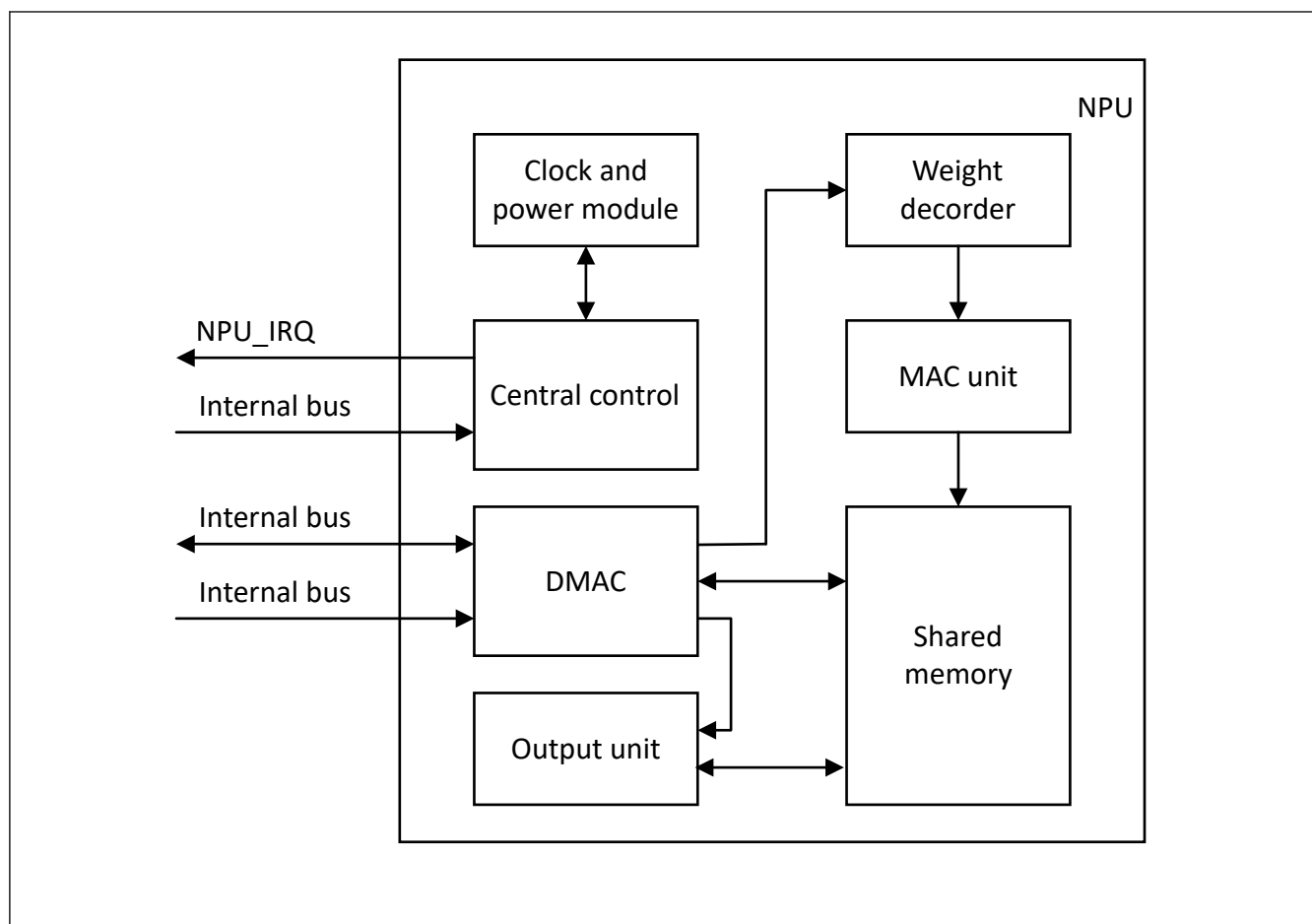


Figure 19.1 NPU block diagram

Please refer to Arm® Ethos™-U55 NPU Arm® Ethos™ Technical Reference Manual for the details.